

11/12/2025 Professor Mukherjee-Yves Greatti
Meeting Minutes

I described the set of models developed for the research project:

- The pre-operative model functions as an initial diagnostic tool that integrates tumor MRI data and pre-operative visual assessments to estimate the likelihood of visual improvement. A multinomial version of this model was also developed to provide a more detailed classification of potential outcomes (*worsened*, *unchanged*, or *improved*).
- A corresponding post-operative model was created to match the pre-operative framework, including surgical variables such as the Endoscopic Endonasal Approach (**EEA**), a minimally invasive surgical technique often used for skull-base and pituitary tumor removals, and Extent of Resection (**EOR**), which indicates how much of the tumor was surgically removed (e.g., gross total resection, subtotal resection, partial resection). This model predicts both the likelihood of visual improvement and the expected visual outcome after surgery.

Professor Mukherjee recommended expanding the outcome prediction to include the *worsened* category. However, because the available dataset is limited ($n = 50$) and the worsened outcome is underrepresented, model performance for this category was notably reduced.